

MANI KETHAN REDDY CHALLA

B.Tech.- Computer Science and Machine Learning

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Summary

Computer Science (AI & ML) graduate with a strong foundation in Python, Java, machine learning, and data structures. Passionate about building AI-powered applications and leveraging AI-assisted development tools to solve real-world problems.

Skills

- Programming Languages: Java, Python, SQL
- AI & Machine Learning: Artificial Intelligence (AI), Machine Learning (ML), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), FAISS
- Tools: Git, GitHub
- Web Technologies: HTML, CSS

Projects

College ChatBot

- Developed using Python, HTML, CSS, Flask, Google API, and prompt engineering at MLRIT Hackathon
- Assists students with college queries using a custom dataset and Google API integration for dynamic responses

Birds Species Detection

- Built an AI-powered bird species detection web application using Python, Flask, React.js, PyTorch, ONNX Runtime, and Hugging Face Transformers for accurate image-based classification.
- Developed a two-stage classification pipeline to reject non-bird images and identify bird species across 525 classes, with real-time species information retrieval through Wikipedia APIs.

AI-Based Content Summarization using RAG

- Built an AI-driven summarization web app using Python, and Retrieval-Augmented Generation (RAG) integrated with Ollama LLM, FAISS, for intelligent context retrieval.
- Designed to analyze and summarize long documents efficiently by combining vector-based retrieval and generative AI, providing concise, accurate outputs in real time.

AlgoSolveo - LeetCode Tutor Chrome Extension

- Built an AI-powered Chrome side panel extension using JavaScript and Chrome APIs to provide interactive, guided tutoring for LeetCode and DSA problems.
- Designed a structured learning roadmap covering 315 DSA problems across 49 patterns, featuring three tutoring styles and step-by-step guidance from brute-force to optimized solutions.

Experience

Intern, DRDO, RCI Government / PSU / Defence	12/2022 – 05/2023 Hyderabad, Telangana, India
• Involved in fabrication and assembly of specialized cables for defense systems • Participated in testing procedures, gaining exposure to quality checks, electrical testing, and validation processes in a real-time defense environment	

Education

B.Tech.- Computer Science and Machine Learning, CMR Technical Campus	08/2023 – 04/2026
Diploma - Electronics and Communication Engineering, TKR College Of Engineering and Technology	09/2020 – 06/2023
SSC, NARAYANA HIGH SCHOOL	06/2019 – 06/2020

Publications

AI-Based Content Summarization using RAG  , Published in IJERT (International Journal of Engineering Research & Technology), Volume 15, Issue 06, June 2026	16/06/2026
• Developed an AI-powered content summarization system using Retrieval-Augmented Generation (RAG). • Integrated FAISS and Large Language Models (LLMs) for efficient semantic document retrieval. • Built the solution using Python to generate accurate, context-aware summaries.	

Certificates

- Salesforce-Agentforce Specialist 

Key Achievements

- First runner-up in Zeneith-Data Voyage, Scope club at MLRIT**
72-hour Hackathon
- First runner-up in AI FUSION FEST, AI and Intel club at CMRTC**
National level 24-hour Virtual Hackathon